LongRange-PFN: KL-Distilled Recursive Latent Summaries for Prior-Fitted Networks beyond the Context Window

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Abstract

Prior-fitted networks (PFNs) return a Bayesian posterior in one forward pass, but only up to the context size seen in training: attention is quadratic and quality drops past that window. LongRange-PFN compresses the distant context D' into k latent tokens $z = \mathrm{rep}(D')$ with a learned summarizer, and predicts from the local context D with z added as extra tokens. We distill a frozen full-context teacher PFN by matching predictive distributions; since both use the same discretized output, the KL is exact. A recursive summarizer $z_t = \mathrm{rep}(C_t, z_{t-1})$ folds the distant context one chunk at a time at constant state, so inference is not capped by the training window. On Gaussian-process priors, where the true posterior is known, the student's NLL falls 0.03–1.30 nats below a lower bound available to any method that sees only D, closing 97% of the truncation gap; and in a streaming setting at $4\times$ the training window the recursive summary beats a teacher restricted to the most recent points, recovering $\approx 39\%$ of the remaining headroom. We also report where it fails: multi-chunk summarization trails single-shot, and bounded-state inference still loses to classical full-context baselines on this easy prior. Against the selectors CRUMB studies, at a matched budget, the summary is closest to the full-context teacher at $n = 16$; per-query nearest neighbors win at $n = 8$; under a disjoint covariate shift the selectors degrade less.

1 Introduction

Prior-fitted networks (Müller et al., 2022) treat Bayesian prediction as in-context learning: pre-train a transformer on data sets drawn from a prior, then at test time feed it a training set D_{tot} and a query x and read off an approximation to $p(y \mid x, D_{\mathrm{tot}})$. TabPFN (Hollmann et al., 2023, 2025) made this a practical tabular model. But the context is the model's memory: every observation is a token, attention is $O(|D_{\mathrm{tot}}|^2)$, and the network never saw contexts larger than some N in training. Past N it faces a cost problem and a distribution shift, and the usual fix, subsampling D_{tot}, discards likelihood terms and provably widens the posterior.

Existing approaches (Section 4) optimize a compact context per data set at test time (Feuer et al., 2024; Ma et al., 2024), retrieve real neighbors (Thomas et al., 2024), or, in language models, compress prompts into soft tokens (Mu et al., 2023; Ge et al., 2024; Chevalier et al., 2023). None gives a PFN a fixed-size, reusable summary of what the far-away data contributed to the posterior.

LongRange-PFN splits $D_{\mathrm{tot}} = D' \cup D$, compresses D' into k latent tokens $z = \mathrm{rep}_\phi(D')$ with a learned summarizer (one forward pass, no test-time optimization), and predicts from $[\, z \mid D \mid x \,]$ with a standard PFN backbone that treats z as ordinary context. Training distills a frozen full-context teacher PFN: teacher and student share the same discretized ("bar") output, so the KL

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between their predictions is an exact categorical KL with no sampling. The summarizer is recursive, $z_t = \mathrm{rep}_\phi(C_t, z_{t-1})$: it folds a stream of chunks into a fixed-size summary, mirroring the Bayesian filtering update $p(\rho \mid C_{1:t}) \propto p(C_t \mid \rho)\, p(\rho \mid C_{1:t-1})$ over a latent task ρ. This decouples how much data the system uses from the window any component was trained on: distilled only on contexts of at most 80 points and 3 chunks, the model still handles 320 points in five chunks at inference, beating a strong recency baseline (Table 3).

Contributions.

1. A summary-conditioned PFN trained by exact predictive KL distillation (Section 3): $\mathrm{rep}_\phi(D')$ is learned end-to-end so that $q_\phi(y \mid x, D, z)$ matches a frozen teacher's $p_\theta(y \mid x, D_{\mathrm{tot}})$, with an exact bucket-level KL.
2. A recursive summary update (Section 3.3) giving bounded-state PFN inference beyond the pre-training window, to our knowledge the first such mechanism for PFNs.
3. Validation against the exact GP posterior (Section 5), including a lower bound the student provably crosses, plus an honest account of what still fails.
4. A matched-budget comparison with uniform subsampling, per-query kNN, MICP, and CRUMB, on the same teacher and prior, including covariate shift (Section 5.1).

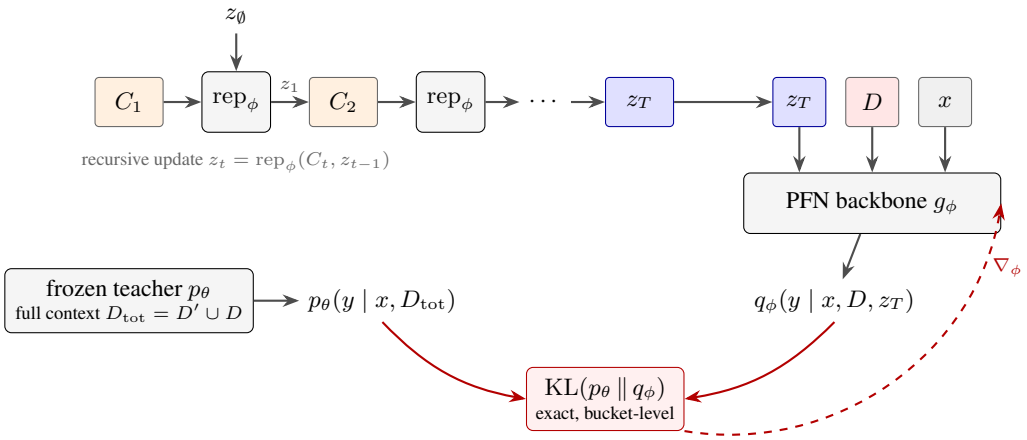


Figure 1: **LongRange-PFN.** D' streams through the summarizer in chunks; the recursive update keeps an $O(k)$ state (left). The backbone predicts from $[\, z_T \mid D \mid x \,]$, treating summary tokens as context (right). Training minimizes the exact bucket-level KL to a frozen full-context teacher (bottom); no ground-truth posterior is needed.

2 Background

PFNs approximate the posterior. Let $p(\rho)$ be a prior over tasks and $p(\mathcal{D} \mid \rho)$ a likelihood over data sets. A PFN p_θ is trained by sampling $\mathcal{D} \sim p(\mathcal{D})$, splitting it into a context D_{tot} and queries (x, y), and minimizing $\mathbb{E}[-\log p_\theta(y \mid x, D_{\mathrm{tot}})]$. Müller et al. (2022) show this minimizes the expected KL to the true posterior predictive up to a constant, so the minimizer is

$$p(y \mid x, D_{\mathrm{tot}}) \;=\; \int p(y \mid x, \rho)\; p(\rho \mid D_{\mathrm{tot}})\, d\rho. \tag{1}$$

For regression, p_θ outputs a *bar distribution*: a piecewise-constant density over B buckets with borders fixed from prior quantiles.

The window problem. Let N be the largest context length seen in training. For $|D_{\mathrm{tot}}| > N$ a PFN faces $O(|D_{\mathrm{tot}}|^2)$ attention and a context size it never saw. Subsampling D_{tot} discards likelihood terms in Eq. (1) and provably widens the posterior; Section 5 measures the cost.

3 LongRange-PFN

3.1 Summary-conditioned prediction

Split $D_{\text{tot}} = D' \cup D$ with $|D'| \gg |D|$. LongRange-PFN has two parts sharing one embedding space:

$$z = \text{rep}_\phi(D') \in \mathbb{R}^{k \times e} \qquad \text{(summarizer)}, \tag{2}$$

$$q_\phi(y \mid x) = g_\phi(x, D, z) \qquad \text{(summary-conditioned PFN)}. \tag{3}$$

The summarizer embeds each pair $(x_i, y_i) \in D'$ with the PFN's encoders, appends k learned query tokens, and runs attention where the queries read the data tokens; the query embeddings are z. The predictor is a standard PFN backbone, initialized from the teacher, run on $[\, z_1 \dots z_k \mid D \mid x_{\text{test}} \,]$ with single_eval_pos $= k + |D|$, so summary tokens sit in the context and are attended to like real samples. When $D_{\text{tot}} = D$, z is a learned null summary $z_\emptyset$ and the model reduces to a plain PFN.

3.2 Predictive KL distillation

Let p_θ be a frozen teacher PFN trained on full contexts. We train ϕ (summarizer and backbone jointly) to minimize

$$\mathcal{L}(\phi) = \mathbb{E}_{D_{\text{tot}} \sim p(\mathcal{D}),\, x}\, \text{KL}\big(\, p_\theta(y \mid x, D' \cup D) \;\big\|\; q_\phi(y \mid x, D, \text{rep}_\phi(D'))\,\big)\,, \tag{4}$$

with $|D|$ from a mixture of small and near-window sizes and, with probability 0.1, $D' = \emptyset$. This is knowledge distillation (Hinton et al., 2015; Korattikara et al., 2015; Malinin et al., 2020) at the level of predictive distributions, but exact: teacher and student share bucket borders, so the KL collapses to a categorical KL over bucket probabilities, $\sum_b p_b \log(p_b/q_b)$, computed directly from the two logit vectors. As a forward KL, it penalizes the student most where it misses teacher mass, pushing errors toward over-coverage rather than overconfidence.

A small value of Eq. (4) does not by itself bound the gap to the true posterior in Eq. (1), the student can only be as good as the teacher, and KL has no triangle inequality, so Section 5 compares against the exact GP posterior, not just the teacher. A fixed-size z works because conditioning matters only through the posterior over the task,

$$p(y \mid x, D', D) = \int p(y \mid x, \rho)\; p(\rho \mid D', D)\, d\rho, \tag{5}$$

so z need only carry what D' contributes to $p(\rho \mid \cdot)$, a learned approximate sufficient statistic (Chen et al., 2021) the predictor reads as a prior that D then updates. This is exact and finite-dimensional for exponential-family tasks; for a GP it is not, but its effective dimension shrinks with the kernel's spectral decay, which is why a small k works while remaining a real bottleneck (Section 6).

3.3 Recursive summaries: streaming past the window

We make rep_ϕ recursive. Split D' into chunks $C_1, \dots, C_T$ and set

$$z_0 = z_\emptyset, \qquad z_t = \text{rep}_\phi(C_t,\, z_{t-1}), \qquad t = 1, \dots, T, \tag{6}$$

prepending the tokens of z_{t-1} to the chunk inside the summarizer. This mirrors Bayesian filtering: z_t plays the role of the running posterior $p(\rho \mid C_{1:t})$, updated with $O((|C_t| + k)^2)$ attention per chunk and $O(k)$ persistent state. Distillation only needs D_{tot} within the teacher's window, but inference can keep folding in new chunks, so the model runs past the window it was trained on. Every prefix z_t is a usable prediction, and raw data can be discarded once folded into z. During distillation we sample $T \in \{1, 2, 3\}$ uniformly, so one set of weights covers single-shot and streaming use (Algorithm 1).

3.4 A certificate for information flowing through the summary

An ablation shows removing z hurts; we can say more. Since expected NLL is minimized by the true conditional, there is a hard lower bound for any method that sees only D.

Proposition 1. *Let (D_{tot}, x, y) be drawn from the prior and let q depend on the data only through (x, D). Then*

$$\mathbb{E}\,[-\log q(y \mid x, D)] \;\geq\; \mathbb{E}\,[-\log p(y \mid x, D)]\,,$$

with $p(y \mid x, D)$ the exact posterior predictive given D alone.

Algorithm 1: Recursive KL distillation of LongRange-PFN

Input: frozen teacher p_θ; prior $p(\mathcal{D})$; window N; bucket borders
initialize backbone of q_ϕ from θ; summarizer encoders from θ;
for *each step* **do**
  sample $\mathcal{D} \sim p(\mathcal{D})$ with N context and M query points;
  sample $|D|$ (or degenerate $D' = \emptyset$); split $D_{\text{tot}} = D' \cup D$; sample $T \in \{1, 2, 3\}$;
  $z \leftarrow z_\emptyset$; **for** $t = 1, \ldots, T$ **do**
    $z \leftarrow \text{rep}_\phi(C_t, z)$
  $\mathcal{L} \leftarrow \frac{1}{M} \sum_x \text{KL}(p_\theta(y \mid x, D_{\text{tot}}) \,\|\, q_\phi(y \mid x, D, z))$; `// exact, bucket-level`
  update ϕ by SGD on $\mathcal{L}$;

Proof. $\mathbb{E}[-\log q] - \mathbb{E}[-\log p] = \mathbb{E}_{x,D} \text{KL}(p(\cdot \mid x, D) \,\|\, q(\cdot \mid x, D)) \geq 0$ by Gibbs' inequality, taking $y \sim p(\cdot \mid x, D)$. □

In our GP setting $p(y \mid x, D)$ has a closed form, so the bound is computable: if a model's test NLL falls below it, that model uses information beyond D, which for our student can only have arrived through z.

Remark 1. *Under exchangeability, chunk order in Eq.* (6) *should not affect the target, but the learned update need not be associative. A regularizer enforcing* $\text{rep}(C_2, \text{rep}(C_1, z_\emptyset)) \approx \text{rep}(C_1 \cup C_2, z_\emptyset)$ *would allow parallel summarization; we leave this for future work.*

4 Related work

Table 1: Positioning. *Amortized*: single forward pass at test time. *Latent*: summary in embedding space. *PPD-distilled*: trained against a frozen Bayesian teacher's predictive distribution. *Recursive*: constant-size state over a stream. *Beyond window*: demonstrated past the backbone's pre-training window.

Method	Amortized	Latent	PPD-distilled	Recursive	Beyond window
TuneTables (Feuer et al., 2024)	✗	✓	✗	✗	✓
ICD (Ma et al., 2024)	✗	✗	✗	✗	✓
LoCalPFN (Thomas et al., 2024)	✓	✗	✗	✗	✓
MICP (Xu et al., 2024)	✓	✗	✗	✗	✓
CRUMB (Heredge et al., 2026)	✗	✗	✗	✗	✓
Gisting / ICAE (Mu et al., 2023; Ge et al., 2024)	✓	✓	✗	✗	✗
AutoCompressors (Chevalier et al., 2023)	✓	✓	✗	✓	✓
Neural Processes (Garnelo et al., 2018)	✓	✓	✗	✗	—
LongRange-PFN (ours)	✓	✓	✓	✓	✓

Scaling PFN context. TuneTables (Feuer et al., 2024) compresses a data set into soft prompts, but per data set by gradient descent; $\text{rep}_\phi(D')$ produces z in one forward pass. ICD (Ma et al., 2024) optimizes a small synthetic set per data set, so the summary lives in data space. LoCalPFN (Thomas et al., 2024) skips compression and retrieves real kNN neighbors. MICP (Xu et al., 2024) clusters the training set and routes queries to a shared support. Closest in goal is CRUMB (Heredge et al., 2026), which selects a distribution-matched real subset per query batch and runs exact PFN inference on it. It keeps real points, needs the test batch up front, and carries no state across batches. Section 5.1 wraps these selectors around our teacher at a matched token budget. TabICL and TabICLv2 (Qu et al., 2025, 2026), TabFlex (Zeng et al., 2025), and Chunked TabPFN (Sergazinov and Yin, 2025) instead change the attention pattern; they still condition on raw tokens and are pretrained for benchmarks such as TabArena (Erickson et al., 2025). None of these trains an encoder against the KL to a full-context teacher, which is what lets LongRange-PFN transfer calibrated uncertainty rather than point accuracy alone.

Context compression in language models. Gisting (Mu et al., 2023), ICAE (Ge et al., 2024), AutoCompressors (Chevalier et al., 2023) (the LM analogue of Eq. (6)), and recurrent memory

transformers (Bulatov et al., 2022) all compress prompts into tokens, but train on next-token likelihood, with no exact distribution-level KL and no posterior to check against.

Distillation, neural processes, sufficient statistics. Distribution-level distillation goes back to Hinton et al. (2015); Bayesian dark knowledge (Korattikara et al., 2015) and ensemble distribution distillation (Malinin et al., 2020) compress model averaging into weights. LongRange-PFN instead compresses conditioning data into a latent, with an exact KL between two bar-distribution PPDs. Neural processes (Garnelo et al., 2018; Kim et al., 2019) also encode a context into a latent, but train from scratch with an ELBO; we distill an already-trained PFN read by an unmodified attention backbone. Treating z as an approximate sufficient statistic connects to Chen et al. (2021) and amortized in-context posterior estimation (Mittal et al., 2025); neither distills a longer-context teacher or uses a recursive update.

5 Experiments

Setup. We build on the official PFN codebase of Müller et al. (2022).[2] The prior is a 1-d GP (RBF kernel, lengthscale 0.1, outputscale 1.0, noise 10^{-2}), chosen so the teacher stays close to the exact posterior at this model scale (a shorter lengthscale 0.03 overwhelmed teacher and summary alike). Between an 80-point window and a 320-point context the exact posterior improves by ≈ 0.061 nats, the headroom the streaming experiment is measured against. The teacher is a 4-layer PFN ($e = 128$, 0.9M parameters, $B = 100$ buckets), trained for $12{,}000$ steps on windows $|D_{\text{tot}}| \leq 80$ with 40 query points. The student (1.6M parameters, with a 3-layer summarizer, $k = 16$ tokens) is initialized from the teacher and distilled for $15{,}000$ steps, sampling $|D|$ from a small ($\mathcal{U}\{5, \ldots, 25\}$) and a near-window regime ($\mathcal{U}\{26, \ldots, 75\}$) with $n_{\text{chunks}} \in \{1, 2, 3\}$. We select the checkpoint by lowest KL on a held-out validation set at $|D| \in \{5, 10, 20, 40\}$ plus the null-summary case (Appendix A). The pipeline runs in under two hours on a laptop GPU.

Baselines. *teacher-full* sees all of D_{tot}; *teacher-local* sees only D (truncation); *GP exact* is the analytic posterior; *no-z* is the student with z replaced by the null token.

Closing the truncation gap. Table 2 and Figure 2 sweep $|D|$ with $|D_{\text{tot}}| = 80$ fixed. Truncation costs up to 1.38 nats of NLL versus teacher-full. The summary recovers almost all of it: at $|D| = 10$ the student closes 97% of the gap, and the single-shot student tracks the full-context reference across the whole sweep. Removing the summary erases the effect, and the no-z curve sits on top of teacher-local. At every $|D|$ the student's NLL falls below the Proposition 1 bound: by 1.30 nats at $|D| = 5$, narrowing to 0.03 nats at $|D| = 40$ as D becomes informative on its own (Figure 4). No method seeing only D can do this in expectation, so D' reaches the predictor through z. On a single draw (Figure 6) the truncated PFN is diffuse away from D while the student sharpens toward the full-context prediction using z alone.

Multi-chunk summarization is worse: splitting the same D' into T recursive chunks costs 0.019 nats of KL at $T = 1$, rising to 0.283 nats at $T = 5$ (Figure 5). The cause is checkpoint selection: the validation set only evaluates $n_{\text{chunks}} = 1$, so nothing rewarded multi-chunk accuracy. The $T = 5$ number still beats an earlier checkpoint's single-shot number (~ 0.73 nats), so the recursion has not regressed; adding multi-chunk splits to validation should close most of the gap.

Table 2: Predictive quality vs. size of the direct context D ($|D_{\text{tot}}| = 80$ fixed, 3,200 data sets, 40 query points each). Left: NLL of held-out targets (nats, lower is better). Right: KL from teacher-full (nats/point).

	NLL (nats)						KL to teacher-full (nats)		
$\|D\|$	GP exact (full)	teacher full	teacher local	LR-PFN	LR-PFN (3 chunks)	no-z (abl.)	teacher local	LR-PFN	no-z (abl.)
5	-0.788 ±0.002	-0.724 ±0.002	0.658 ±0.006	**-0.698** ±0.003	-0.506 ±0.003	0.720 ±0.008	1.334 ±0.005	**0.022** ±0.000	1.399 ±0.007
10	-0.788 ±0.002	-0.724 ±0.002	0.141 ±0.006	**-0.702** ±0.003	-0.535 ±0.003	0.171 ±0.007	0.830 ±0.005	**0.020** ±0.000	0.864 ±0.006
20	-0.788 ±0.002	-0.724 ±0.002	-0.349 ±0.004	**-0.708** ±0.003	-0.604 ±0.003	-0.345 ±0.005	0.357 ±0.003	**0.016** ±0.000	0.364 ±0.004
40	-0.788 ±0.002	-0.724 ±0.002	-0.614 ±0.003	**-0.714** ±0.002	-0.664 ±0.002	-0.616 ±0.003	0.105 ±0.001	**0.011** ±0.000	0.105 ±0.001

[2] `https://github.com/automl/PFNs`

Streaming beyond the teacher's window. Table 3 and Figure 3 test $|D_{\text{tot}}|$ up to 320, four times the teacher's window, under a matched budget: the student keeps the 70 most recent points as D and folds everything older into z recursively (chunks ≤ 70), while the teacher uses the 80 most recent points, the best any window-limited PFN can do. Both attend over the same number of tokens; the only difference is the $O(k)$ latent. The student's NLL improves monotonically as the stream grows (five recursive steps at $|D_{\text{tot}}| = 320$, where training used three), while the window teacher is flat by construction. From $|D_{\text{tot}}| = 160$ onward the student beats the window teacher, recovering 39% of the headroom between the windowed and full-stream exact posteriors. At $|D_{\text{tot}}| = 80$, where one chunk covers all of D', student and window teacher tie; the advantage appears once the stream needs more than one chunk, exactly when the recency baseline starts discarding data the summary keeps.

The student's state is $k \times e$ floats regardless of stream length, while a recency window must grow or slide. That storage argument is asymptotic, but at our scale 2,048 floats only beat raw points past $\sim 10^3$ observations, so the win at $|D_{\text{tot}}| = 320$ is architectural, not yet a memory saving. What the experiment shows is that discarding raw data costs nothing in accuracy here (relevant when data cannot be retained, e.g. privacy) and that every prefix z_t answers mid-stream. Two caveats: the comparison is against a recency window, not unbounded full-context inference (Appendix B shows classical methods still ahead of both); and it uses the checkpoint whose multi-chunk quality we flagged (Figure 5), so the recursion's ceiling is likely higher than Table 3 shows.

Table 3: Streaming beyond the window: test NLL (nats) as $|D_{\text{tot}}|$ grows, with matched budgets: student: 70 recent points + recursive z (chunks ≤ 70); teacher: the 80 most recent points (1,280 data sets per cell). LongRange-PFN overtakes the window teacher from $|D_{\text{tot}}| = 160$ onward.

$\lvert D_{\text{tot}}\rvert$	GP exact (full)	GP exact (window)	teacher (window)	LR-PFN (stream)
80	-0.786 ±0.004	-0.786 ±0.004	-0.724 ±0.004	**-0.722** ±0.004
160	-0.835 ±0.003	-0.787 ±0.003	-0.727 ±0.004	**-0.746** ±0.004
240	-0.854 ±0.003	-0.792 ±0.004	-0.731 ±0.004	**-0.755** ±0.004
320	-0.860 ±0.003	-0.788 ±0.003	-0.727 ±0.004	**-0.754** ±0.003

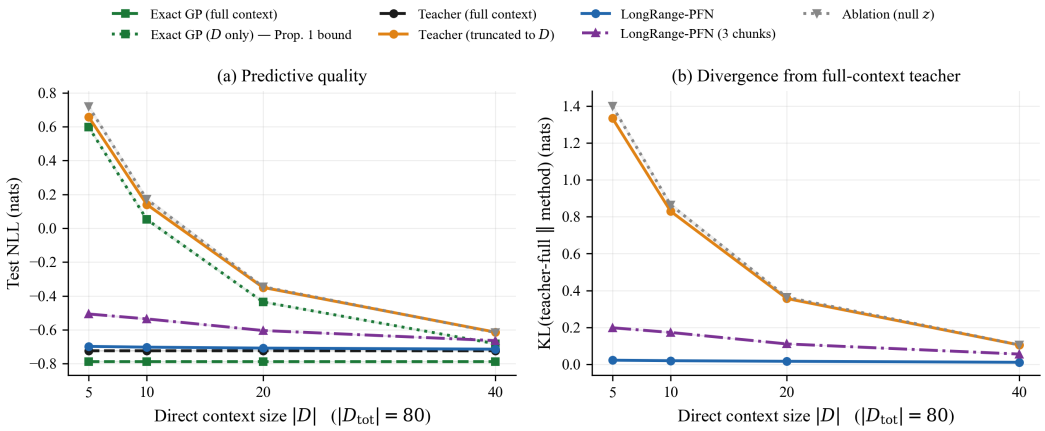


Figure 2: **Closing the truncation gap** (mean ± 95% CI, 3,200 data sets). (a) Test NLL vs. $|D|$: the single-shot student (blue) tracks the full-context reference (black, dashed) and stays below the exact-GP-on-D bound (green, dotted); the 3-chunk variant (purple) trails. (b) KL to teacher-full; the no-z ablation (gray) tracks truncation.

Calibration and where the summary helps. Coverage of central predictive intervals at $|D| = 10$ (Figure 7, left) shows the student's reliability curve closest to the diagonal, with truncation under-covering and teacher-full mildly conservative, consistent with the forward KL preserving the teacher's uncertainty. Binning test points by distance to the nearest point of D (right) shows the student's gain over truncation growing with distance and vanishing near D, the spatial signature of the certificate. A linear probe of z (Appendix C) decodes the full-context predictive mean from $\text{vec}(z)$ across the domain (held-out $R^2 \in [0.3, 0.9]$), so z encodes the function rather than memorizing points.

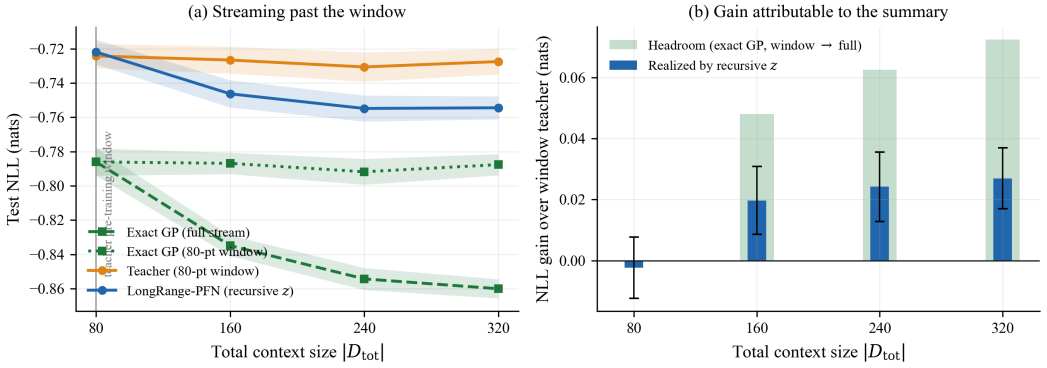


Figure 3: **Streaming beyond the window** (mean $\pm$ 95% CI, 1,280 data sets). (a) Test NLL vs. total context size: the window teacher (orange) is flat, the exact GP on the full stream (green, dashed) improves with evidence, and the recursive student (blue) surpasses the teacher as the stream grows. (b) Gain over the window teacher (blue) against the headroom to the full exact posterior (green).

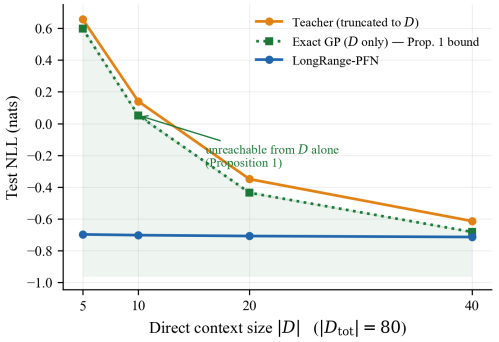


Figure 4: **The Proposition 1 certificate.** The exact GP on D (green, dotted) lower-bounds any D-only method; the shaded region is unreachable without outside information. The student (blue) sits inside it at every $|D|$.

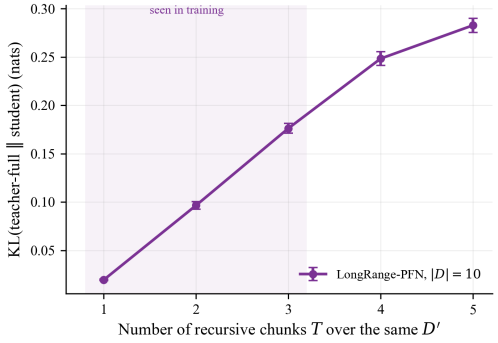


Figure 5: **Recursion cost.** KL to teacher-full when the same D' ($|D'| = 70$, $|D| = 10$) is folded in T chunks: 0.019 nats at $T = 1$ to 0.283 nats at $T = 5$. Shaded: training range ($T \leq 3$). Traced to a validation criterion that only checks $T = 1$ (Appendix A).

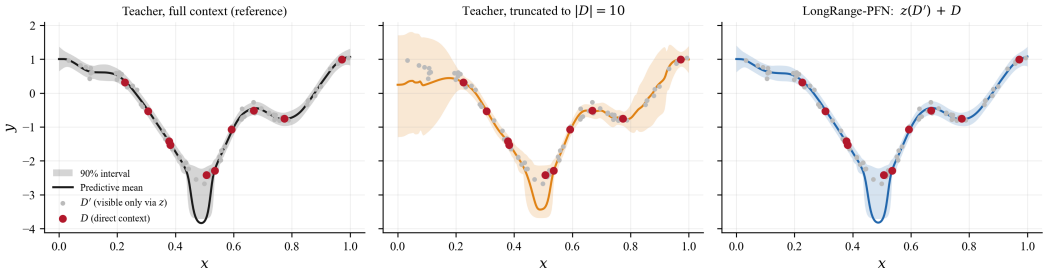


Figure 6: **Qualitative behaviour** on one GP draw ($|D| = 10$, red; D', gray, visible to the student only through z). Predictive mean and 90% interval. The truncated PFN (middle) is overdispersed away from D; LongRange-PFN (right) sharpens toward the full-context predictive (left) using z alone.

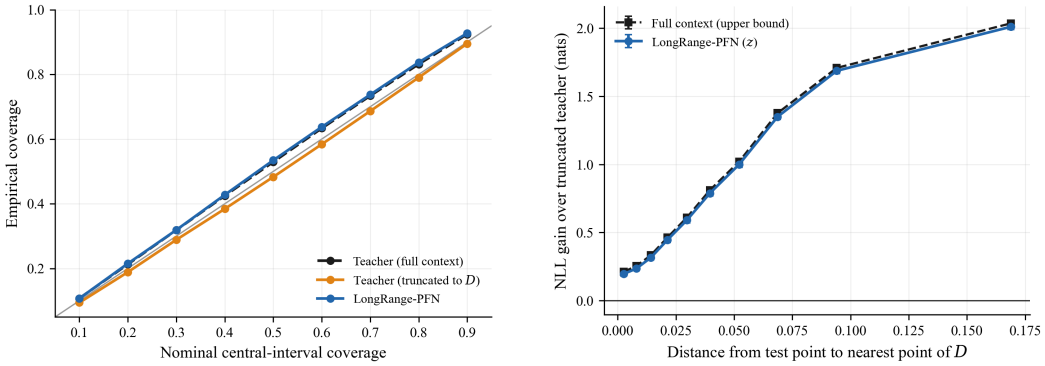


Figure 7: **Diagnostics at $|D| = 10$.** Left: coverage of central predictive intervals; the student (blue) is closest to the diagonal. Right: NLL gain over teacher-local, binned by distance to the nearest point of D; the gain grows with distance and stays below the full-context ceiling (dashed).

5.1 Context selection at a matched budget

CRUMB (Heredge et al., 2026) reports accuracy and RMSE on TabArena (Erickson et al., 2025) using TabPFNv2, TabICLv1, and TabICLv2. LongRange-PFN is a 1-d GP student, so we reimplement CRUMB's four selectors — uniform subsampling, per-query kNN, MICP (Xu et al., 2024), and CRUMB — and wrap them around our frozen teacher. MICP uses $\lceil N/n \rceil$ training clusters. CRUMB uses $K = 8$ test clusters and the greedy MMD step of their Equation (3), with the median-heuristic RBF bandwidth. Standardization uses training statistics. Selectors attend n raw tokens. The student attends n raw tokens plus $k = 16$ summary tokens: either a random window, or the $K = 1$ MMD subset, with the complement summarized (chunks of 70 past that size).

Table 4: Matched budget inside the window ($N = 80$, 128 data sets, 40 queries). Selectors run the teacher on n real points. LongRange-PFN adds a summary of the complement. KL is to the teacher on all 80 points. Bold is the lowest mean among budget-n methods. Mean ± s.e.

Method	NLL (nats) $n = 8$	$n = 16$	$n = 40$	KL to teacher-full $n = 8$	$n = 16$	$n = 40$
Teacher, full context	-0.720 ±0.012	-0.720 ±0.012	-0.720 ±0.012	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000
Uniform	0.299 ±0.026	-0.168 ±0.026	-0.619 ±0.013	0.966 ±0.024	0.528 ±0.021	0.099 ±0.006
Per-query kNN	**-0.646 ±0.013**	-0.685 ±0.012	-0.708 ±0.012	**0.066 ±0.002**	0.030 ±0.001	0.007 ±0.001
MICP	-0.577 ±0.012	-0.634 ±0.013	-0.690 ±0.012	0.126 ±0.003	0.079 ±0.004	0.022 ±0.002
CRUMB	-0.599 ±0.012	-0.683 ±0.012	**-0.713 ±0.012**	0.112 ±0.003	0.031 ±0.001	**0.004 ±0.000**
LongRange-PFN	-0.630 ±0.012	**-0.706 ±0.012**	-0.710 ±0.012	0.087 ±0.005	**0.018 ±0.001**	0.011 ±0.001
LongRange-PFN + MMD window	-0.638 ±0.012	-0.703 ±0.012	-0.711 ±0.012	0.083 ±0.006	0.019 ±0.001	0.012 ±0.001

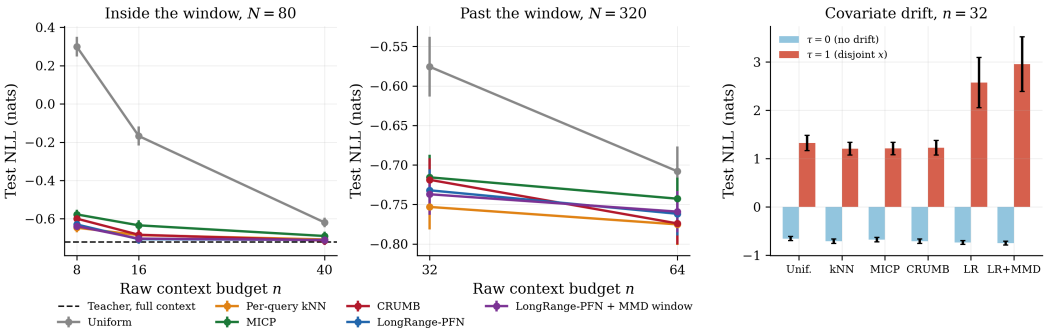


Figure 8: **Selectors versus the latent summary** (mean ± 95% CI). Left: the student meets the full-context teacher by $n = 16$, while uniform subsampling stays far above it. Middle: at $N = 320$, per-query kNN is lowest. Right: under a disjoint shift ($\tau = 1$) the summary variants degrade past uniform subsampling.

On $N = 80$ (Table 4, Figure 8, left), uniform subsampling is far from the teacher at $n = 8$ (KL 0.966) and $n = 16$ (KL 0.528). At $n = 8$, kNN is closest (KL 0.066); the student (KL 0.087) beats CRUMB (KL 0.112) on 66% of data sets and beats kNN on 44%. At $n = 16$ the student is closest (KL 0.018, against 0.030 for kNN and 0.031 for CRUMB), with lower NLL than kNN on 67% of data sets and than CRUMB on 70%. At $n = 40$ the NLL values of CRUMB, kNN, and both student variants lie within one standard error.

Past the window ($N = 320$, 64 data sets) the teacher is not run on the full pool. At $n = 32$, kNN reaches -0.753, the MMD-window student -0.737, the random-window student -0.732, and CRUMB -0.719. At $n = 64$, kNN (-0.775) and CRUMB (-0.774) tie, and the student (-0.762) is about 0.01 nats behind, consistent with the multi-chunk cost in Figure 5.

Table 5: Covariate drift along x, budget $n = 32$, training pool from the lower half of x, capped at 160 points (40 data sets). $\tau = 0$ tests inside that half; $\tau = 1$ tests on the disjoint upper half. Mean $\pm$ s.e.

Method	$\tau = 0$	$\tau = 1$
Uniform	-0.655 $\pm$0.022	1.323 $\pm$0.080
Per-query kNN	-0.709 $\pm$0.023	**1.204 $\pm$0.066**
MICP	-0.679 $\pm$0.023	1.210 $\pm$0.065
CRUMB	-0.708 $\pm$0.023	1.226 $\pm$0.076
LongRange-PFN	-0.738 $\pm$0.021	2.575 $\pm$0.264
LongRange-PFN + MMD window	**-0.749 $\pm$0.020**	2.956 $\pm$0.289

Under the shift protocol of CRUMB's Section 5.4 (Table 5), $\tau = 0$ favors the summary: the MMD-window student (-0.749) and the random-window student (-0.738) beat kNN and CRUMB. At $\tau = 1$, kNN, MICP, and CRUMB land together near NLL 1.204, while the student rises to 2.575 and the MMD-window student to 2.956, both worse than uniform (1.323). The summary was trained on exchangeable data from one region; on a disjoint test set it conditions on points the queries do not resemble. Test-side retrieval is the right tool for that shift. The recursive summary remains the method that runs when the stream arrives before the queries, which CRUMB cannot do.

6 Limitations

This is a proof of concept on a 1-d GP prior with a small teacher and exchangeable contexts. Chunk order does not matter here but would under non-stationary data, where the recursion should be a proper filter. The summary size k is an information bottleneck we have not characterized as a rate–distortion trade-off. The distillation ceiling is the teacher's own quality within its window; past that the recursion is validated only empirically. Checkpoint selection is a concrete gap: the validation set only checked single-chunk quality, so the selected checkpoint is good at $T = 1$ and worse at higher T (Figure 5), a cheap fix we have not made. Finally, Appendix B shows a fitted classical GP or full-context TabPFN still beat bounded-state inference here, because exact inference on this prior is cheap. A real advantage needs a prior where exact inference is intractable, a teacher at TabPFN's scale, query-aware summaries $z(x)$, and the associativity regularizer of Section 3.3. Under a disjoint covariate shift the summary also degrades by more than the selectors in Section 5.1.

7 Conclusion

LongRange-PFN carries out-of-window evidence in a few latent tokens, trained only by distilling a frozen teacher's predictive distribution. On GP priors the summary provably carries posterior information, crossing the exact lower bound of Proposition 1 at every context size tested, and the recursive update handles streams four times the training window at constant state while beating a strong recency baseline. At a matched budget the summary is closer to the full-context teacher than kNN, MICP, or CRUMB at $n = 16$; per-query kNN wins at $n = 8$, and under a disjoint covariate shift the selectors degrade less (Section 5.1). Two further gaps remain, both concrete and measurable: multi-chunk summarization lags single-shot, and bounded-state inference still trails classical full-context methods on this easy prior.

Reproducibility statement

All code (built on the official PFN repository), training scripts, seeds and configurations, and the scripts that regenerate every number, table and figure from `results/metrics.json` are in the accompanying repository.

A Training curves

Figure 9 shows the optimization traces for both stages. Every regime flattens well before the final step, which is why we select the reported checkpoint by validation KL rather than by step count, and why the single-shot-only validation set leaves multi-chunk quality unoptimized.

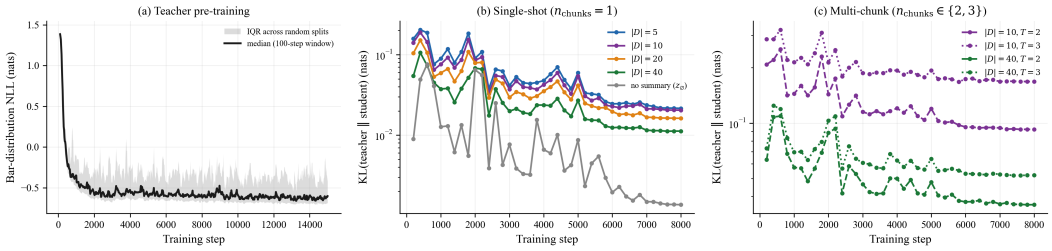


Figure 9: **Optimization traces.** (a) Teacher pre-training: bar-distribution NLL (rolling median, solid; IQR, shaded). (b) Student distillation: validation KL(teacher ‖ student), one line per $|D|$ regime. Every regime flattens by step $\sim 12{,}000$, so we select by validation KL rather than final step. This validation set only covers single-shot summarization, which is why the checkpoint is strong at $T = 1$ and worse at higher T (Figure 5).

B Defensive baselines at large context

The main text asks whether the summary preserves information truncation destroys. A harder question: why not give all the data to a method with no window? TabPFN, a fitted GP, and a random forest have no 80-point limit, so here we give them the entire context, up to $8\times$ the teacher's window, and check whether LongRange-PFN, a recent local window plus an $O(k)$ recursive summary, holds up.

Table 6 covers $N \in \{80, 160, 320, 640\}$ using an earlier teacher/student pair trained with a shorter schedule; the conclusion follows from the prior being easy for a matched kernel, not from the checkpoint, but we flag the mismatch. Full-context baselines are TabPFN (Hollmann et al., 2025), a scikit-learn GP with an RBF kernel fit per data set (unlike the "GP exact" oracle, which uses the true hyperparameters), kernel ridge, k-NN, and a random forest, all given the full N points; LongRange-PFN and the recency teacher see only bounded state.

A fitted RBF GP ends up close to the oracle, unsurprising since the data come from an RBF GP, and full-context TabPFN is also strong; both beat the bounded-state methods in NLL by 0.1–0.2 nats. This is an easy point-prediction task for a matched kernel, so it was never going to show a predictive win over classical methods. That is fine: the contribution is not a better predictor on GP data, where exact inference is cheap, but a bounded-state mechanism for carrying evidence when the full context cannot be stored and re-attended. A predictive win over strong full-data baselines would need a prior where exact inference is intractable and a teacher at TabPFN's scale.

C What does the summary encode?

We probe what information the latent z carries with a ridge decoder from $\mathrm{vec}(z)$ to the full-context predictive mean and a PCA of the summary space (Figure 10).

Table 6: Large-context defensive baselines over total context size N (up to $8\times$ the teacher's window; 100 data sets per cell, TabPFN on its own smaller sample). *Full-context* baselines see all N points; *bounded-state* methods see only recent points plus, for LongRange-PFN, the $O(k)$ summary. NLL (nats) for probabilistic methods, RMSE for all; mean $\pm$ s.e.

NLL (nats, lower is better)				
Method	$N = 80$	$N = 160$	$N = 320$	$N = 640$
Teacher (80-pt window)	-0.693 $\pm$0.016	-0.682 $\pm$0.015	-0.697 $\pm$0.014	-0.716 $\pm$0.014
LongRange-PFN (bounded)	-0.678 $\pm$0.016	-0.662 $\pm$0.014	-0.684 $\pm$0.014	-0.702 $\pm$0.012
GP exact (full, oracle)	-0.789 $\pm$0.012	-0.824 $\pm$0.011	-0.859 $\pm$0.011	-0.893 $\pm$0.010
TabPFN (full context)	-0.632 $\pm$0.024	-0.723 $\pm$0.020	-0.782 $\pm$0.020	-0.756 $\pm$0.024
GP-RBF fitted (full)	-0.771 $\pm$0.014	-0.815 $\pm$0.011	-0.856 $\pm$0.011	-0.891 $\pm$0.011
RMSE (lower is better)				
Method	$N = 80$	$N = 160$	$N = 320$	$N = 640$
Teacher (80-pt window)	0.142 $\pm$0.008	0.154 $\pm$0.009	0.148 $\pm$0.009	0.140 $\pm$0.008
LongRange-PFN (bounded)	0.141 $\pm$0.008	0.153 $\pm$0.008	0.147 $\pm$0.009	0.139 $\pm$0.008
GP exact (full, oracle)	–	–	–	–
TabPFN (full context)	0.120 $\pm$0.004	0.106 $\pm$0.002	0.103 $\pm$0.002	0.101 $\pm$0.002
GP-RBF fitted (full)	0.111 $\pm$0.002	0.106 $\pm$0.001	0.102 $\pm$0.001	0.099 $\pm$0.001
Kernel ridge (full)	0.110 $\pm$0.002	0.106 $\pm$0.001	0.102 $\pm$0.001	0.098 $\pm$0.001
k-NN (full)	0.164 $\pm$0.004	0.125 $\pm$0.002	0.113 $\pm$0.001	0.109 $\pm$0.001
Random forest (full)	0.142 $\pm$0.003	0.127 $\pm$0.002	0.123 $\pm$0.001	0.121 $\pm$0.001

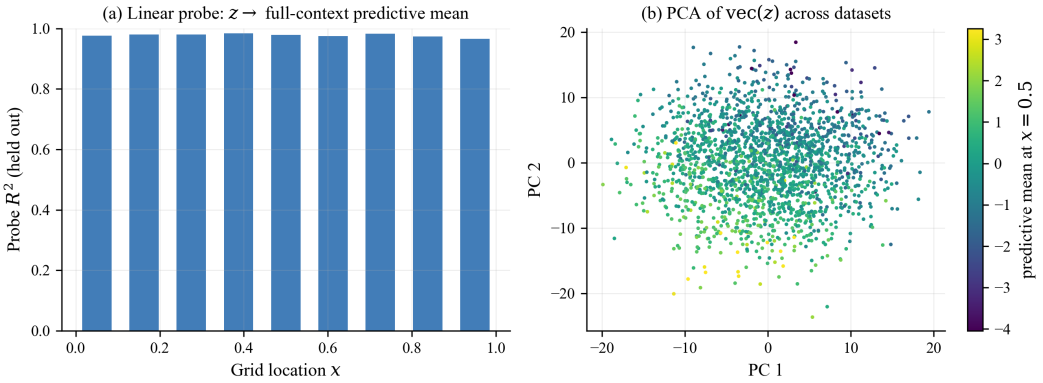


Figure 10: **Probing the summary space** (z from all 80 context points; $\approx$ 1,900 held-out data sets). (a) Held-out R^2 of a ridge probe from $\mathrm{vec}(z)$ to the full-context teacher's predictive mean at each grid location: partially decodable everywhere ($R^2 \approx 0.3$ mid-domain to 0.9 near the boundary). (b) First two principal components of $\mathrm{vec}(z)$, colored by the predictive mean at $x = 0.5$: the space is coarsely organized by function value.

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